

Rank by scalarized reward (1 = best)

	pinned	LOSO deployed					LOLO deployed					LOSO nested					LOLO nested				
MAPPO	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	3	3	1	
IPPO	2	2	2	2	2	2	2	2	3	2	2	2	2	2	2	2	2	2	2	2	
Centralized PPO	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	
Hysteresis (fixed)	3	3	3	3	3	3	3	3	4	3	3	3	3	3	3	3	3	4	4	3	
Always-DPDK	4	4	4	4	4	4	4	4	1	4	4	4	4	4	4	4	4	1	1	4	

Rank by energy (1 = lowest Wh)

MAPPO	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3	2	1	
IPPO	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	2	
Centralized PPO	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	
Hysteresis (fixed)	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	1	1	3	
Always-DPDK	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	

Rank by QoS-violation rate (1 = lowest)

MAPPO	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	3	2	
IPPO	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	2	2	3	
Centralized PPO	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	
Hysteresis (fixed)	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	
Always-DPDK	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
P		0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4

calibration world (P = pinned; digit = fold; columns 12–21 = secondary nested refits)